

Matplotlib Tasks-1

📌 TASK SET 1: Single Line Plot (Very Easy)

Task 1 — Student Performance Line

- Create `x = [1, 2, 3, 4, 5]` (tests)
 - Create `y = [45, 50, 60, 70, 80]` (marks)
 - Plot a line graph
 - Use:
 - `marker='o'`
 - `color='blue'`
 - `linestyle='dashed'`
 - Add title, x-label, y-label, grid
-

Task 2 — Temperature Change

- Create temperature values for 7 days
 - Rotate x-axis labels by **30 degrees**
 - Use `marker='s'`
 - Show grid
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📌 TASK SET 2: Random Data (Fun + Visual)

Task 3 — Random Growth Graph

- Generate **20 random values** using `np.random.randint`
- Plot them
- Use:
 - `color='green'`
 - `linestyle='dotted'`
 - `marker='o'`

☞ Concept reinforced: `np.random`, simple plotting

Task 4 — Random X vs Random Y

- Generate:
 - `x = np.random.randint(1, 50, size=15)`
 - `y = np.random.randint(1, 100, size=15)`
 - Plot them
 - Enable grid
 - Add proper labels
-

☐ TASK SET 3: Multiple Line Plot (Most Important)

Task 5 — Two Students Marks

- Student A marks → one line
 - Student B marks → second line
 - Use:
 - different colors
 - different markers
 - Add `legend()`
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Task 6 — Company Sales Comparison

- `x = [2019, 2020, 2021, 2022, 2023]`
 - Company A sales → red line
 - Company B sales → green line
 - Legend position: `upper left`
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☐ TASK SET 4: Styling Practice (Very Easy)

Task 7 — Marker Change

Plot the same data **3 times**, changing only:

- `marker (o, s, ^)`
- `color`

Task 8 — Grid ON / OFF

- Plot a line
- First show **without grid**
- Then show **with grid**

□ TASK SET 5: Error Finding

Task 9 — Find the Mistake

```
x = [1,2,3,4]
y = [10,20,30]
```

```
plt.plot(x,y)
plt.show()
```

☞ Students must explain **why error occurs**
